BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 119 · Lesson 119 · Spiral Development and Method Selection

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is one distinguishing XP practice?

Q2. An OR gate has inputs 0 and 1. What is the output?

Q3. What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

Q4. What is the time complexity of sequential search in the worst case?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

spiral model — Risk-driven iterative development lifecycle.

risk analysis — Estimating likelihood and impact of potential problems.

prototype — Early model built to explore or test an idea.

iteration — Repeated cycle of development and evaluation.

New learning

Plan the cycle — Set objectives, options and constraints for a spiral iteration.

Analyse risks — Evaluate and reduce technical or organisational uncertainties before commitment.

Prototype and review — Develop/verify increment, consult stakeholders, plan next risk-driven cycle.

Worked example

A bank is building a novel payment verification system with security uncertainty. Justify the spiral model.

1. Model answer — Each cycle sets goals, identifies risks such as fraud or performance, prototypes/assesses the riskiest elements, evaluates with stakeholders then plans the next cycle. Risk reduction drives the project. — Check: link each claim to the scenario and use precise OCR terminology.

We Do · Practise with guidance

A museum app needs a basic pilot within weeks but must support disabled visitors. Select a method.

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

RAD/agile could produce accessible prototypes quickly with visitor feedback, but test accessibility systematically. Spiral may be excessive for a small low-risk app; compare project scale and uncertainty.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Describe three central activities within one spiral development cycle. [6]

Question 2

Give two reasons a fixed-requirement small project might not benefit from the spiral model. [6]

Question 3

Compare waterfall, agile/RAD, XP and spiral for a high-risk changing specification. [8]

Exit Ticket · Answer before checking

Spiral development is simply waterfall repeated without evaluating project risks.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. For example TDD, pair programming, CI or frequent refactoring.

Q2. 1.

Q3. X is removed; Y remains at the front.

Q4. O(n).

You Do pages · Independent questions

Q1. Objectives planning [2], explicit risk analysis/prototyping [2], development/evaluation/next-cycle planning [2].

Q2. Risk analysis overhead/cost [3]; complex management and limited benefit when uncertainty is low [3].

Q3. At least three accurate contrasting features [4], context-linked risk/testing/cost discussion [2], justified selection or combination [2].

Exit Ticket · Model answer

False. Risk identification and reduction drive the cycles.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: